Rajagopal Somaskandan

Chennai, India | *+91 7200288242* | *rajagopal-somaskandan.netlify.app* | *somaskandan931@gmail.com* | *linkedin.com/in/somaskandan-rajagopal* | *github.com/Somaskandan931*

Professional Summary

Applied Machine Learning Engineer and IEEE-published researcher (M.Sc. CS, 2027) with experience building **production-grade AI systems**. Delivered high-impact models including fraud detection (94% recall, ROC-AUC 0.995) and real-time ML systems, with end-to-end ownership from training to FastAPI deployment.

Technical Skills

Languages: Python, Java, SQL
Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, SHAP
NLP & GenAI: BERT, Sentence-Transformers, spaCy, NLTK, LangChain, ChromaDB, FAISS, RAG
Computer Vision & 3D: OpenCV, Optical Flow, COLMAP, 3D Gaussian Splatting, FFmpeg
Databases: PostgreSQL, MongoDB, Redis, SQLite
Frameworks & Tools: FastAPI, Flask, Streamlit, Docker, Git, GitHub, Netlify, Render
Data Analysis: Pandas, NumPy, Matplotlib, Plotly, Power BI

Education

Sathyabama Institute of Science and Technology — **B.Sc. Computer Science (AI Specialization)** | First Class — *2025*
Sathyabama Institute of Science and Technology — **M.Sc. Computer Science** | — *Expected 2027*

Experience

Production AI Developer (Intern) | *InnoXR Labs Pvt. Ltd.* — *Mar 2026 – Jun 2026*
- Developed and deployed ML-powered FastAPI microservices for production AR/immersive applications, enabling scalable API-first AI integration.
- Owned end-to-end feature delivery — from model development to API deployment — contributing directly to production releases in a fast-paced startup environment.

Data Analyst Intern | *India Advocacy* — *Jul 2024 – Oct 2024*
- Built Java + Selenium pipelines to scrape and process ***1M+ records***, transforming raw data into ***15–20k high-quality datasets*** through multi-stage validation.

Security Trooper (National Service) | *Singapore Armed Forces* — *Jan 2020 – Jan 2022*
- Supported security operations, operated authorized transport vehicles under regulated operational environments and assisted Ministry of Health COVID-19 task force initiatives through structured data collection and reporting

Sales Associate (Part-Time) | *Bata* — *Feb 2022 – Aug 2022*
- Managed high-volume customer interactions, delivering efficient service in a fast-paced retail environment.
- Handled billing, inventory coordination, and issue resolution, improving in-store operational efficiency.

Projects

FinGuard Pro: Explainable AI for Financial Fraud Detection — *Research Project (IEEE Publication, ICAAIC 2025)* — *2025*
XGBoost, SHAP, FastAPI, SQL
- Achieved ***94% recall*** and ***ROC-AUC 0.995*** on imbalanced UPI-style fraud data; published at ICAAIC 2025 (ISBN: 979-8-3315-6587-9).
- Layered rule-based AML heuristics (velocity abuse, structuring patterns) on XGBoost to cut false positives; SHAP outputs satisfy regulatory audit requirements.

CareerGenie: AI-Powered Job Matching & Skill Gap Analysis — *Hackathon Winner* — *2026*
Claude API, Sentence-Transformers, ChromaDB, FastAPI, React
- Built RAG pipeline (MiniLM-L6-v2 + ChromaDB) for semantic resume–job matching with ***granular 0–100% scoring***; Claude API generates per-role skill-gap reports.
- Won ***Best Industry Innovation Award*** at national-level Industry Innovation ’26 (ATRIBS + Bluewind Innovations, 63+ teams).

MonoSplat: Single-Video 3D Gaussian Splat Reconstruction — *Internship Project (InnoXR Labs)* — *2026*
PyTorch, COLMAP, FFmpeg, FastAPI, Three.js
- End-to-end pipeline: single phone video → photorealistic browser-navigable 3D scene; no depth sensors or multi-camera rig required.
- CPU/GPU split — COLMAP SfM locally, custom `GaussianTrainer` on Colab T4 (L1 + SSIM loss, adaptive densification); Three.js export at ***30+ FPS***.

S.A.F.E: Real-Time Crowd Anomaly & Risk Detection — *Design thinking Project* *2026*

OpenCV, Farneback Optical Flow, MAD, FastAPI, Streamlit

- Designed real-time, privacy-preserving anomaly detection API using zone-based optical flow aggregation; streams live CCTV input without storing raw footage.
- Reduced false positives by ***40–60%*** via global motion compensation for camera shake — eliminated the critical failure mode blocking drone and handheld deployment.
- Achieved ***ROC-AUC 0.8391*** on cross-domain Mall dataset with ***zero retraining*** — architectural decision for field portability over single-dataset tuning.

Redline: Multilingual Preventive Healthcare Chatbot — *SIH 2025* *2026*

IndicBERT, Groq LLaMA 3.3, FastAPI, Telegram Bot API

- Fine-tuned IndicBERT across 10+ Indian languages; ***82.4% intent accuracy*** + ***95.6% safety filter accuracy*** via confidence-aware LLM fallback routing.
- Live Telegram deployment: 3-tier medical escalation + real-time outbreak alerts across all 28 Indian states.

SourceUp: Constraint-Aware Explainable Supplier Recommendation — *Final Year Capstone* *2026*

Sentence-BERT, FAISS, FastAPI, Scikit-learn

- Constraint-aware ranking (budget, delivery, reliability) over Sentence-BERT + FAISS semantic retrieval; explainability layer surfaces per-feature trade-off insights for procurement decisions.

Publications & Conferences

FinGuard Pro: Explainable AI for Financial Fraud Detection | *ICAAIC 2025* *ISBN: 979-8-3315-6587-9*

Real-Time Explainable Fake News Detection using BERT-LSTM | *ICASET 2025*

Achievements & Hackathons

Industry Innovation Hackathon | *Sathyabama Institute of Science and Technology — National Level* *Mar 2026*

- Won ***Best Industry Innovation Award*** at Industry Innovation ’26 (organized with ATRIBS and Bluewind Innovations) — defeating 63+ teams from institutions across India with the CareerGenie AI platform.

Origin 24-Hour Hackathon | *SIMATS Engineering, Chennai* *Apr 2026*

- Competed in a 24-hour sprint organized by the SIMATS Hackathon Club, Saveetha Institute of Medical and Technical Sciences.